

Program : Diploma in Polymer Technology	
Course Code : 5093C	Course Title: Chemical Engineering
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4,T:0,P:0)	Periods per semester: 60

Course Objectives:

- To understand the basic units, unit operations & unit process in chemical engineering.
- To comprehend the mechanism of heat transfer by conduction, convection, radiation and evaporation.
- To explain and differentiate the mechanism of mass transfer by diffusion, absorption, leaching, extraction, adsorption and drying.
- To describe the basic principles and types of distillation.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics in Chemical Engineering		Applied Physics I	1
Conduction, Convection, Radiation, Absorption, Adsorption, Leaching, Extraction etc.		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Identify the basic units, unit operations & unit process in chemical engineering.	14	Applying
CO2	Apply mechanism of heat transfer by conduction, convection, radiation to solve problems in chemical engineering.	15	Applying
CO3	Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, extraction, adsorption and drying.	14	Understanding

CO4	Generalize the basic principles and types of distillation	15	Applying
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the basic units, unit operations & unit process in chemical engineering.		
M1.01	Explain the various systems of units	3	Understanding
M1.02	Describe unit operations and unit process in the field of chemical engineering	3	Understanding
M1.03	Solve problems related to atomic weight, molecular weight and equivalent weight of a solid substance.	4	Applying
M1.04	Solve problems related to expressing concentration of a solution and gases.	4	Applying
Contents: Units and Dimensions - various systems of Units - Fundamental & Derived Units- concept of unit operations and unit process - gm atom, kg atom, gm mole, kg mole - Atomic weight, Molecular weight, Equivalent Weight - normality - molarity - molality - mass percent - Mass, Volume, Mole relation, - relationships for Gaseous Systems - simple problems.			
CO2	Apply mechanism of heat transfer by conduction, convection, radiation to solve problems in chemical engineering.		
M2.01	Identify different modes of heat transfer	3	Understanding

M2.02	Apply Fourier's law of thermal conduction to derive heat transfer equations for various solid surfaces and solve problems related to it.	8	Applying
M2.03	Illustrate various heat exchangers used in industry	2	Understanding
M2.04	Discuss the types of evaporators used in industry.	2	Understanding
	Series Test - I	1	
Contents: Conduction - Fourier's law - derivation - problems - Conduction through Plane wall, Cylindrical wall - derivations - problems -, Convection - concept of natural convection, Radiation idea of black body - gray body - emissivity - absorptivity, Heat exchanger patterns - Shell and Tube - Double Pipe Heat Exchangers – Evaporators - horizontal and vertical evaporators.			
CO3	Paraphrase mechanism of mass transfer by diffusion, absorption, leaching, extraction, adsorption and drying.		
M3.01	Discuss gas absorption and gas absorption equipment's.	4	Understanding
M3.02	Generalize leaching technique	3	Understanding
M3.03	Describe extraction and extraction equipment's	3	Understanding
M3.04	Summarize drying equipment's used in industry	4	Understanding
Contents: Diffusion - Fick's law - law of conservation of mass - Henry's law - gas absorption - working of packed tower - flooding - channeling - leaching - applications - working of Bollman Extractor - Extraction - raffinate, extract - working of mixer settler - adsorption - physical and chemical adsorption - Drying - applications - working of tray and spray drier			
CO4	Generalize the basic principles and types of distillation		
M4.01	Differentiate various types of distillation.	3	Understanding
M4.02	Apply Raoul's law and Daltons law to derive equation connecting mole fraction and vapor of a binary mixtures	5	Applying
M4.03	Derive equation for relative volatility	4	Applying
M4.04	Compare Azeotropic and Extractive Distillation	3	Understanding
	Series Test - II	1	
Contents: Distillation - types - Simple and Steam distillation -fractional distillation - Raoult's law - daltons law - Relative volatility - derivation - simple problems - Minimum and Maximum boiling Azeotropes -comparison - Azeotropic and Extractive Distillation.			

Text / Reference:

T/R	Book Title/Author
T1	Process Calculations for chemical engineers - Chem. Engg., Edn Development Centre, IIT Madras
R1	Process Calculations - V. Venkataramani, N. Anantharaman
R2	Introduction to Chemical Engineering - Walter.L.Badger (Publisher: McGraw-Hill Inc. US; International 1955 Edition)
R3	Unit Operations - McCabe SMITH (Publisher - McGraw-Hill Education; 7edition)
R4	Mass Transfer - Robert E Treybal (Publisher - McGraw-Hill Book Company; 3rd edition)

Online Resources:

Sl.No	Website Link
1	https://www.researchgate.net
2	http://cc.sit.edu.cn
3	http://www.chem.ucalgar